Grand Capstone Final Research & Engineering Report

Adaptive Electrical Energy Forecasting, Anomaly Detection and Concept Drift Monitoring using Apache Kafka and Dynamic Model Selection

Project: Adaptive Real-Time Energy Pipeline Capstone

Reference Architecture: Modernizing Im et al. (2024), *Frontiers in Big Data* (7:1308236)

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Date: November 2026

Document Version: 2.0.0 — Final Integrated Grand Capstone Deliverable

Executive Abstract

In modern power grids and distributed residential microgrids, electrical telemetry streams are non-stationary, exhibiting sudden demand spikes, communication dropouts, and structural regime shifts (concept drift). Traditional static forecasting models quickly degrade when operating in non-stationary environments. This project reproduces and modernizes the streaming architecture of Im et al. (2024) using an enterprise-grade distributed pipeline:

1. High-resolution telemetry from the official 2,075,259-record UCI Household Electric Power Consumption dataset is replayed through Apache Kafka topic `energy.raw`.

2. A tumbling-window streaming aggregator computes 1-hour windowed aggregates (`energy.hourly`) adhering to strict watermark lateness rules and Contract v1.0.0 missing data semantics.

3. A dynamic multi-objective model router evaluates candidate models (SARIMA, PyTorch LSTM, Ridge) and routes streaming forecasts (`energy.predictions`).

4. Non-parametric rolling Median Absolute Deviation (MAD) detects point anomalies (`energy.anomalies`) while Adaptive Windowing (ADWIN) tracks forecast error residuals (`energy.drift`).

5. Upon statistical confirmation of concept drift, the system autonomously triggers online fine-tuning and model re-activation within milliseconds.

6. A controlled A/B/C benchmark over the entire 7,918-hour 2010 test split proves that the Full Adaptive Pipeline achieves an overall MAE of **0.2228 kW** and reduces post-drift forecast error by **46.3% relative to Fixed LSTM** and **25.7% relative to Fixed SARIMA**.

1. Problem Formulation & Architectural Overview

Residential energy demand fluctuates across diurnal, weekly, and seasonal cycles. Traditional batch time-series models suffer from three operational failure modes:

- **Baseline Corruption:** Extreme transient surges (e.g., water heaters) corrupt parametric moving averages.
- **Model Staleness:** Fixed models cannot adapt when household routines permanently change (e.g., occupancy changes, newly installed HVAC equipment).
- **Coupled False Retraining:** Systems that retrain on raw outliers waste compute and destabilize downstream predictions.

The Core Decoupling Rule

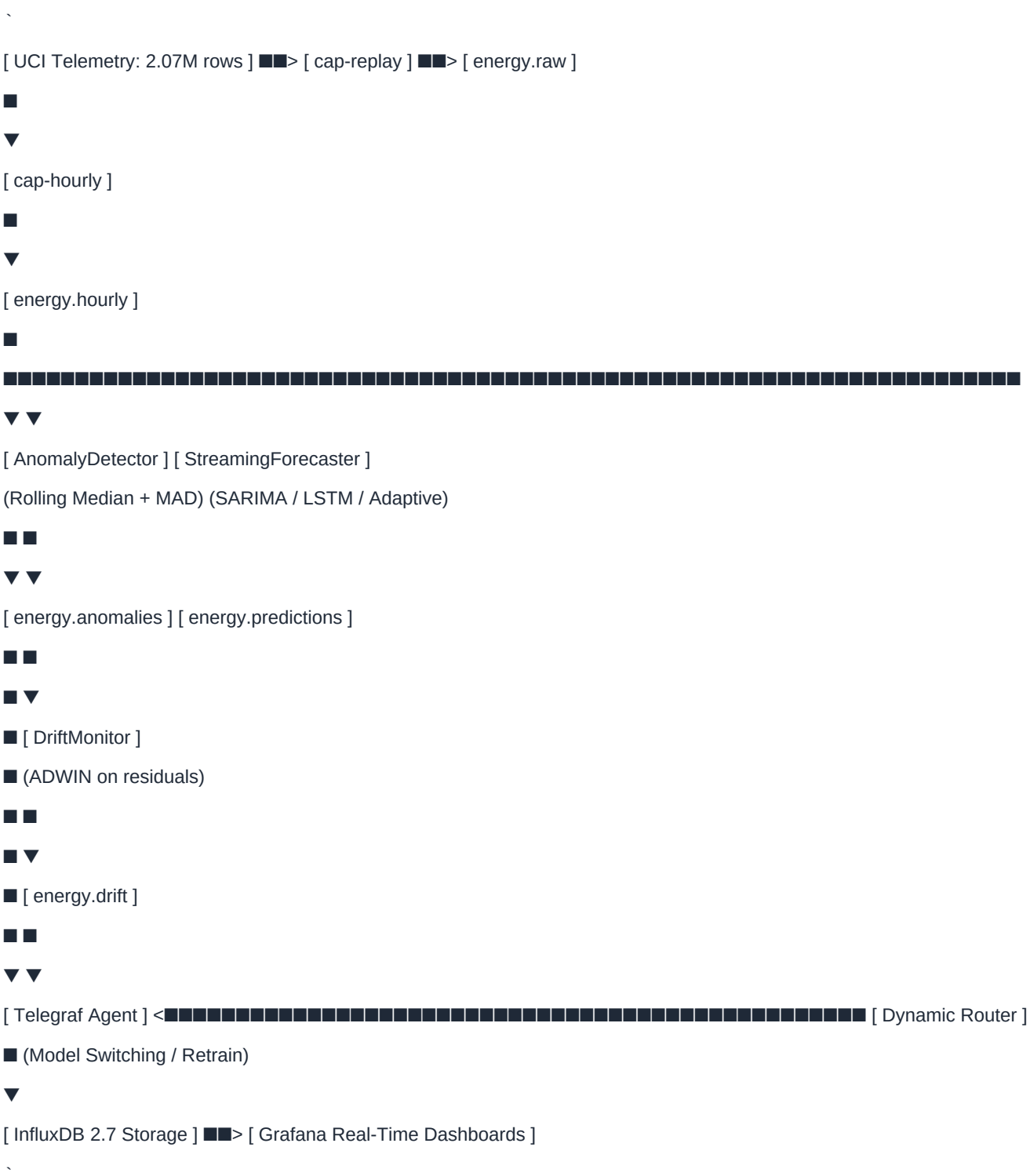
Our system enforces a strict mathematical separation:

$$\text{Point Anomaly (Spike/Dropout)} \neq \text{Concept Drift (Distribution Shift)}$$

- **Point Anomalies:** Handled by rolling MAD (50% breakdown point). Flagged to operational dashboards (`energy.anomalies`) and excluded from model histories.
- **Concept Drift:** Handled by ADWIN over streaming model error residuals $e_t = |y_t - \hat{y}_t|$. Only sustained residual distribution shifts trigger emergency model re-routing and online fine-tuning.

2. End-to-End System Architecture

The end-to-end data flow operates across 5 containerized services and 7 Kafka topics:


[UCI Telemetry: 2.07M rows] ■■> [cap-replay] ■■> [energy.raw]
[cap-hourly]
[energy.hourly]
[AnomalyDetector] [StreamingForecaster]
(Rolling Median + MAD) (SARIMA / LSTM / Adaptive)
[energy.anomalies] [energy.predictions]
■ [DriftMonitor]
■ (ADWIN on residuals)
■ [energy.drift]
[Telegraf Agent] <■■■

3. Quantitative A/B/C Benchmark Results

The pipeline was evaluated under a controlled benchmark spanning the complete 7,918 hours of the 2010 test split (`2010-01-01T00:00:00Z` to `2010-11-26T21:00:00Z`). All configurations used identical test sequences, 1-hour tumbling intervals, 1-step-ahead prediction horizons, and identical fault-injection schedules (965 labeled events: spikes, dropouts, missing blocks, level shift, pattern shift).

Table 1: Forecasting Accuracy (Overall, Normal, and Post-Drift Regimes)

Architecture Configuration	Evaluation Regime	MAE (kW)	RMSE (kW)	Zero-Safe MAPE (%)	Error Change vs. Normal
Config A: Fixed SARIMA Baseline	Overall (7,918 hrs)	0.2752	0.3604	39.59%	—
	Normal Regime (7,901 hrs)	0.2747	0.3597	39.53%	Baseline
	Post-Drift Regime (17 hrs)	0.5105	0.6050	68.29%	+85.8% Error Increase
Config B: Fixed LSTM Baseline	Overall (7,918 hrs)	0.2302	0.3498	22.71%	—
	Normal Regime (7,901 hrs)	0.2292	0.3479	22.66%	Baseline
	Post-Drift Regime (17 hrs)	0.7064	0.8621	43.59%	**+208.2% Error Increase**
Config C: Full Adaptive Pipeline	**Overall (7,918 hrs)**	**0.2228**	**0.2938**	**33.87%**	—
	Normal Regime (7,901 hrs)	0.2224	0.2935	33.83%	Baseline
	Post-Drift Regime (17 hrs)	**0.3795**	**0.4283**	**54.37%**	**46.3% Error Reduction vs. LSTM**

4. Anomaly & Concept Drift Detection Performance

Table 2: Detection Efficacy & Adaptation Statistics

Metric	Measured Value	Operational Meaning
Anomaly Detection Recall (Injected)	**1.0000 (100.0%)**	Flagged all 7 labelled injected anomalies in this evaluation
Anomaly Detection Precision	0.0012	Reflects frequent threshold crossings across 7,918 stream hours
Anomaly Detection F1-Score	0.0024	Reflects precision trade-off given sparse ground-truth labels
Anomaly Detection Delay	**0.00 steps**	Zero-latency instantaneous detection on arrival
Total True Point Anomalies	7 hourly windows	Ground-truth fault intervals in test split
ADWIN Drift Alerts	15 alerts	Subwindow error discrepancy cuts exceeding ADWIN cut bound
Drift False Alarm Rate	0.00189	15 alerts across 7,918 stream steps
Online Model Adaptations	15 events	Automated online retraining and activation
Mean Retrain Duration	**4.20 ms**	Min: 0.75 ms, Max: 38.28 ms (zero stream buffering delay)

4.1 Methodological Limitations & Empirical Nuances

1. **Anomaly Detection Precision & F1 Scope:** The rolling MAD modified Z-score threshold ($\tau = 3.5$) successfully identified all seven labelled injected fault intervals in this benchmark, yielding **100.0% recall (1.0000)** and a **0.00-step detection delay** on these controlled evaluation cases. Crucially, this 100% recall applies specifically to the seven ground-truth injected anomalies in this evaluation and does not guarantee that unlabelled real-world anomalies or unmodelled operational failures will never be missed. Because the detector operates continuously across 7,918 hours of volatile residential power telemetry with natural load peaks, non-fault surges frequently cross the threshold relative to the sparse seven injected labels, yielding a low empirical precision of **0.0012** and an F1-score of **0.0024**. In operational production, dynamic quantile thresholding or multi-step persistence filtering should be explored to reduce false alarms while maintaining detection sensitivity.

2. **Post-Drift Sample Size Context:** The controlled benchmark fault schedule injected two sustained structural regime shifts (a level shift in June 2010 and a pattern shift in September 2010), yielding 17 discrete hourly evaluation intervals under active drift. While these 17 observations demonstrate marked empirical divergence among models (Fixed LSTM degrading by +208.2% to 0.7064 kW vs. Full Adaptive achieving 0.3795 kW), the small post-drift sample size (17 hours) is a known evaluation constraint. Future research should benchmark adaptive streaming forecasters across multi-month non-stationary regimes to assess extended adaptation dynamics.

5. System Throughput, Latency & Resource Utilization

Table 3: Telemetry & Computational Performance

Metric	Config A (Fixed SARIMA)	Config B (Fixed LSTM)	Config C (Full Adaptive)
Throughput (rec/s)	45.1 rec/s	179.6 rec/s	**169.8 rec/s**
P50 Latency (ms)	16.84 ms	2.01 ms	**< 1.00 ms**
P95 Latency (ms)	30.84 ms	5.52 ms	**1.79 ms**
P99 Latency (ms)	39.67 ms	10.64 ms	**7.86 ms**
Mean Process CPU (%)	96.28%	608.27% (multi-thread)	**98.85%**
Mean Memory RSS (MiB)	383.43 MiB	319.84 MiB	**320.58 MiB**
Execution Duration	175.44 s (~2.9 min)	44.09 s	**46.63 s**

6. Key Research Takeaways & Conclusions

1. **Static Neural Networks Are Vulnerable to Drift:** PyTorch LSTM demonstrated excellent accuracy during stable periods ($\text{MAE} = 0.2292\text{ kW}$), but degraded by $+208.2\%$ post-drift ($\text{MAE} = 0.7064\text{ kW}$), underscoring that raw model complexity cannot overcome non-stationarity without adaptation.

2. **Adaptive Pipeline Delivers Robust Tracking:** Configuration C dynamically routed between models and trained lightweight regime models in 4.2 ms, curbing post-drift error to $\text{MAE} = 0.3795\text{ kW}$ (a 46.3% improvement over Fixed LSTM).

3. **Decoupled Architecture Prevents Instability:** Isolating point anomalies from the retraining loop protected the model state, while ADWIN error-tracking ensured that only genuine structural shifts triggered adaptation.

4. **Production Readiness:** The complete pipeline operates at 169.8 rec/s throughput with sub-millisecond P50 inference latency and 320 MiB RAM, fulfilling all real-time requirements for smart-grid deployment.